



## Fort Street High School

NESA Number:

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**2024**

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

# Mathematics Extension 1

### General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided
- Marks may be deducted for careless or badly arranged work.
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks : 70    Section I – 10 marks**

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

### **Section II – 60 marks**

- Allow about 1 hours and 45 minutes for this section
- Write your student number on each answer booklet.
- Attempt Questions 11 – 15

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

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1. Given  $f(x) = 1 + \sqrt{x}$ , the domain and range of  $f^{-1}(x)$  are:

- A. Domain:  $\{x \geq 0\}$ , Range:  $\{y \geq 0\}$
- B. Domain:  $\{x \geq 0\}$ , Range:  $\{y \geq 1\}$
- C. Domain:  $\{x \geq 1\}$ , Range:  $\{y \geq 0\}$
- D. Domain:  $\{x \geq 1\}$ , Range:  $\{y \geq 1\}$

2. The acute angle (to the nearest degree) between the vectors  $\underline{u} = \begin{pmatrix} \sqrt{3} \\ 2 \end{pmatrix}$  and  $\underline{v} = \begin{pmatrix} 9 \\ \sqrt{17} \end{pmatrix}$  is:

- A.  $24^\circ$
- B.  $51^\circ$
- C.  $46^\circ$
- D.  $48^\circ$

3. The solution to the inequality  $\frac{3}{x-2} \leq 4$  is given by:

- A.  $x < -2$  and  $x \geq -\frac{11}{4}$
- B.  $x > -2$  and  $x \leq -\frac{11}{4}$
- C.  $x < 2$  and  $x \geq \frac{11}{4}$
- D.  $x > 2$  and  $x \leq \frac{11}{4}$

4. A function is defined as  $f(x) = \tan^{-1}(\tan(x))$ .

The value of  $f\left(\frac{9\pi}{4}\right)$  is:

- A.  $\frac{\pi}{4}$
- B.  $\frac{5\pi}{4}$
- C.  $\frac{7\pi}{4}$
- D.  $\frac{9\pi}{4}$

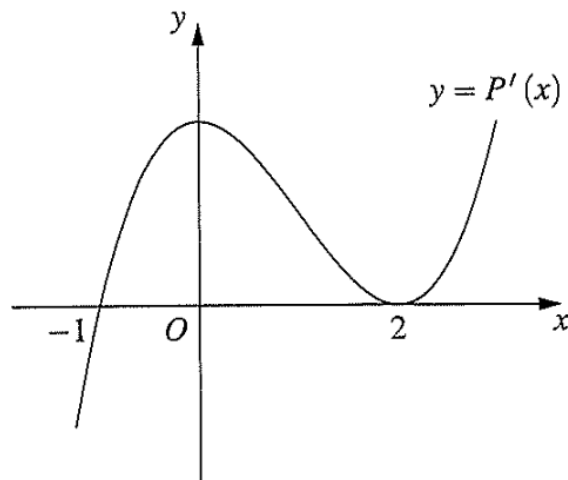
5. When the polynomial  $P(x)$  is divided by  $x+1$  it gives a remainder of  $n^2$ , where  $n$  is a positive integer. When  $P(x)$  is divided by  $(x+1)(x+3)$  it gives a remainder  $(nx+6)$ . The value of  $n$  is:

- A. 1
- B. 3
- C. 6
- D. 2

6. The  $\int \sin 3x \sin x \, dx$  is equal to:

- A.  $\frac{1}{4} \sin 2x - \frac{1}{8} \sin 4x + C$
- B.  $\frac{1}{8} \sin 4x - \frac{1}{4} \sin 2x + C$
- C.  $\frac{1}{4} \cos 2x - \frac{1}{8} \cos 4x + C$
- D.  $\frac{1}{8} \cos 4x - \frac{1}{4} \cos 2x + C$

7. In the graph below,  $y = P'(x)$  represents the first derivative of a polynomial  $P(x)$  of degree 4, where  $P(x)$  has a multiple root.

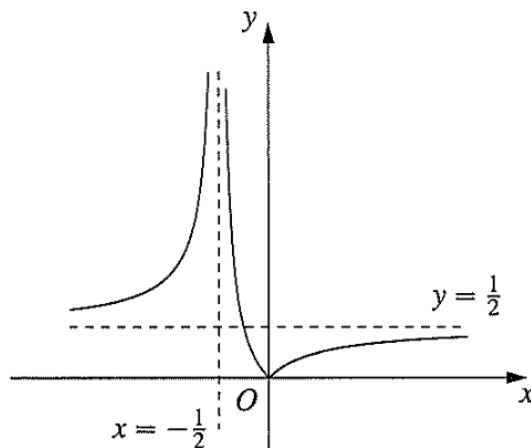


Which of the following could be true about the polynomial  $P(x)$  ?

- A.  $x = -1$  is a root of multiplicity 3.
  - B.  $x = 0$  is a root of multiplicity 2.
  - C.  $x = 2$  is a root of multiplicity 2.
  - D.  $x = 2$  is a root of multiplicity 3.
8. The derivative of  $f(x) = 2x^2 \cos^{-1}(2x)$  is:

- A.  $\frac{-8x}{\sqrt{1-2x^2}}$
- B.  $\frac{-8x}{\sqrt{1-4x^2}}$
- C.  $\frac{-4x^2}{\sqrt{1-2x^2}} + 4x \cos^{-1} 2x$
- D.  $\frac{-4x^2}{\sqrt{1-4x^2}} + 4x \cos^{-1} 2x$

9. The diagram below shows the graph of the function  $y = g(x)$ , which is the result of a set of transformations on the graph of  $y = f(x)$ .



If  $f(x) = \frac{1}{x-2} + 2$ , which equation best represents  $g(x)$ ?

- A.  $g(x) = \frac{1}{|f(x-2)|}$
- B.  $g(x) = |f(x-2)|$
- C.  $g(x) = \frac{1}{|f(x+2)|}$
- D.  $g(x) = |f(x+2)|$
10. Vector  $\vec{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$  is projected on to vector  $\vec{b} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$ . Given that  $\text{Proj}_{\vec{b}} \vec{a} = \frac{1}{2} \vec{b}$ , which of the following could possibly be vector  $\vec{a}$ ?

- A.  $\vec{a} = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$
- B.  $\vec{a} = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$
- C.  $\vec{a} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$
- D.  $\vec{a} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

## Section II

**60 marks**

**Attempt Questions 11 – 15**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 11 – 15, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (13 marks) Use the Question 11 Writing Booklet

- (a) Let  $P$  be the point  $(12, 5)$ . Find the position vector of point  $P$ , and hence find a unit vector in the direction of the position vector. 2
- (b) Find the exact value of  $\int \sin^2 \frac{2}{3}x \, dx$ . 1
- (c) Use the  $t$ -identities to solve the equation  $2 \sin x + \cos x + 1 = 0$  for  $0 \leq x \leq 2\pi$ . 4
- (d) Find  $\int \frac{1}{\sqrt{64 - 49x^2}} \, dx$ . 2
- (e) (i) Express  $\cos x - \sqrt{3} \sin x$  in the form  $R \cos(x + \alpha)$  for  $R > 0$  and  $\alpha$  acute. 2
- (ii) Hence, solve  $\cos x - \sqrt{3} \sin x = 2$  for  $0 \leq x \leq 2\pi$ . 2

**End of Question 11**

**Question 12** (12 marks) Use the Question 12 Writing Booklet

- (a) There are four teams consisting of two tennis players in each team. 2

They are seated around a circular table.

In how many ways can the teams be arranged around the table if the players in each team are to stay together.

- (b) By using the substitution  $x = \sin \theta$ , find  $\int_0^{\frac{1}{2}} (1-x^2)^{-\frac{3}{2}} dx$  3

- (c) Prove by mathematical induction that  $3^{3n-1} + 5^{3n-2}$  is divisible by 7, 4

for any integer  $n \geq 1$ .

- (d) A cake tin cools such that its temperature  $T$  degrees,  $t$  minutes after it is removed 3

from the oven is given by:

$$T = R + Ae^{-kt} \text{ Where } k, R \text{ and } A \text{ are constants}$$

It is known that:

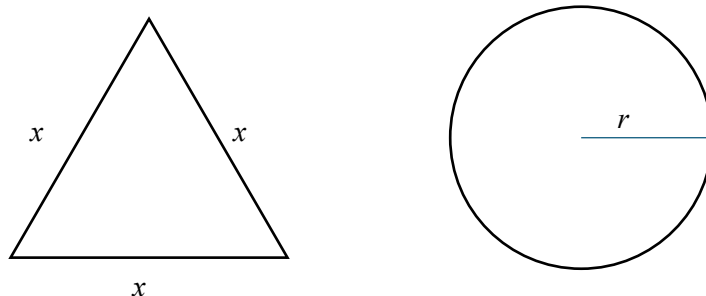
$$\frac{dT}{dt} = -k(T - R) \text{ Do NOT prove this.}$$

When removed from an oven the cake tin has a temperature of  $180^\circ\text{C}$ . If the cake tin takes one minute to cool to  $150^\circ\text{C}$  and the room temperature is  $20^\circ\text{C}$ , find the time to the nearest minute it takes for the cake tin to cool to  $80^\circ\text{C}$ . (Assume that the cake tin cools at a rate proportional to the difference between the temperature of the cake tin and the temperature of the surrounding air.)

**End of Question 12**

**Question 13** (13 marks) Use the Question 13 Writing Booklet

- (a) The side lengths of an equilateral triangle are shrinking at a rate of  $\sqrt{x} \text{ m/s}$ , where  $x$  metres is the length of each side of the triangle. The triangle and the circle shown below always have the same area  $A$ .



- (i) Show that the rate of change of the area of the triangle is:

2

$$\frac{-\sqrt{3} x \sqrt{x}}{2} \text{ m}^2/\text{s}$$

- (ii) Find the rate of change of the radius of the circle when  $x = 5$  metres.

3

Give the answer correct to 2 decimal places.

- (b) By considering  $(1+x)^{2n} + (1-x)^{2n}$ , where  $n$  is a positive integer, show that:

3

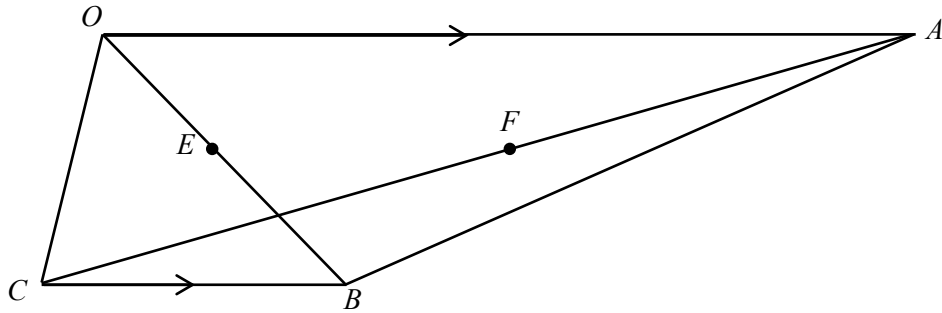
$$1 + \binom{2n}{2} + \binom{2n}{4} + \dots + \binom{2n}{2n} = \frac{4^n}{2}.$$

**Question 13 continues on the next page**

**Question 13 continued**

(c)  $OABC$  is trapezium with  $\overrightarrow{OA} = 3\overrightarrow{CB}$ .  $\overrightarrow{OA} = a$  and  $\overrightarrow{OC} = c$ .

$E$  and  $F$  are midpoints of  $OB$  and  $CA$  respectively.



- |       |                                                                              |          |
|-------|------------------------------------------------------------------------------|----------|
| (i)   | Show that $\overrightarrow{OE} = \frac{1}{2}\left(c + \frac{1}{3}a\right)$ . | <b>1</b> |
| (ii)  | Find $\overrightarrow{EF}$ .                                                 | <b>3</b> |
| (iii) | Hence, prove that $CEFB$ is a parallelogram.                                 | <b>1</b> |

**End of Question 13**

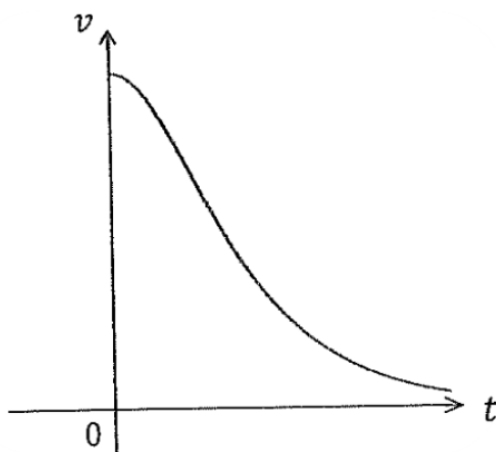
**Question 14** (11 marks) Use the Question 14 Writing Booklet

- (a) A particle  $P$  is moving along the  $x$  axis. At time  $t = 0$ , it was at  $x = 0$ . Its velocity

4

$v \text{ ms}^{-1}$  at a time  $t$  is given by  $v = \frac{e^t}{1 + (e^t)^2}$ .

The graph below shows its velocity as a function of time.



Given that the displacement of the particle when  $t = k$  seconds is  $\tan^{-1} \frac{1}{2}$  metres,

show that  $k = \ln 3$ .

**Question 14 continues on the next page.**

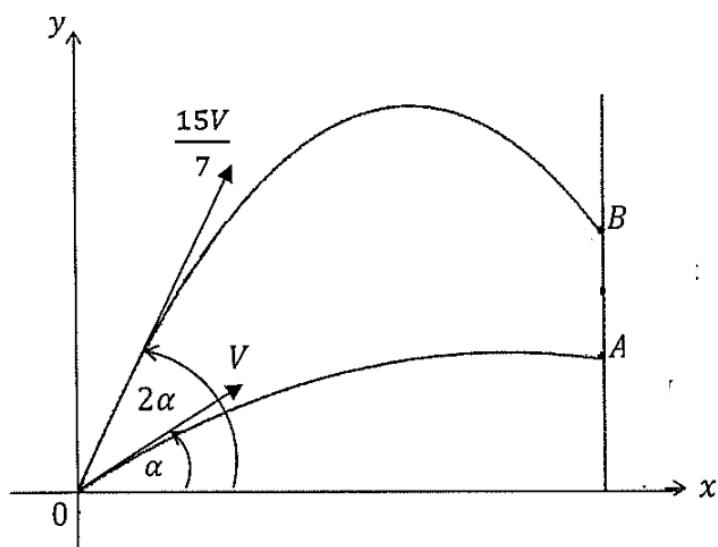
**Question 14 continued**

- (b) Jacob hits two balls at the same time from the origin  $O$  towards a target on a vertical wall.

The velocity of the first ball is  $V \text{ ms}^{-1}$  and its angle of projection is  $\alpha$  to the horizontal

while the velocity of the second ball is  $\frac{15V}{7} \text{ ms}^{-1}$  and its angle of projection to

the horizontal is  $2\alpha$ .



- (i) Show that the position vector of the second ball  $t$  seconds after being projected is given by:

**3**

$$\mathbf{r}_2(t) = \begin{pmatrix} \frac{15V}{7}t \cos 2\alpha \\ -\frac{1}{2}gt^2 + \frac{15V}{7}t \sin 2\alpha \end{pmatrix}$$

- (ii) It is known that both balls hit the wall after  $T$  seconds.

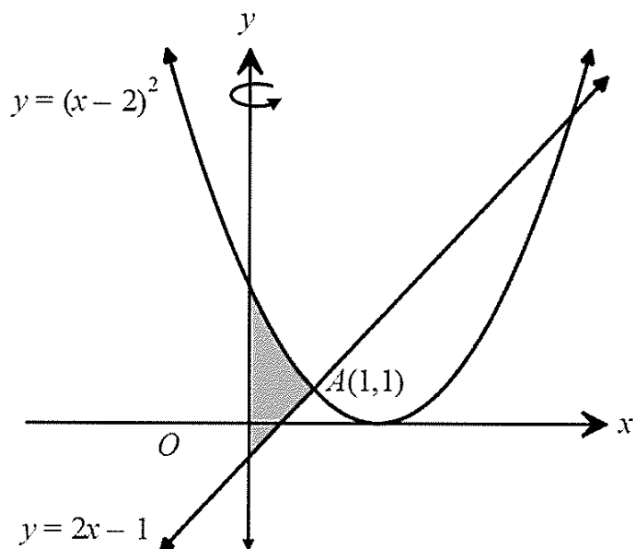
**4**

Show that  $\cos \alpha = \frac{5}{6}$ .

**End of Question 14**

**Question 15** (11 marks) Use the Question 15 Writing Booklet

- (a) Consider the shaded region bounded by the line  $y = 2x - 1$ , the curve  $y = (x - 2)^2$  and the  $y$ -axis. 4



Find the volume of the solid of revolution formed by rotating this region about the  $y$ -axis.

- (b) (i) Show that the solution of  $\cos 3x - \sin 2x = 0$ , for  $0 < x < \frac{\pi}{2}$  is given by 3

$$\sin x = \frac{\sqrt{5} - 1}{4}.$$

You may use the identity  $\cos 3x = 4\cos^3 x - 3\cos x$ . DO NOT PROVE THIS.

- (ii) Without a calculator, verify that  $x = \frac{\pi}{10}$  is a solution to  $\cos 3x = \sin 2x$ . 1

- (iii) Using the results obtained in parts (i) and (ii) prove that 3

$$\sin \frac{\pi}{5} \cos \frac{\pi}{10} = \frac{\sqrt{5}}{4}.$$

**END OF EXAMINATION**



**Fort Street  
High School**

**Solutions**

**2024**

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

# Mathematics Extension 1

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## **Section II – 60 marks**

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## Section I

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A. Domain:  $\{x \geq 0\}$ , Range:  $\{y \geq 0\}$

B. Domain:  $\{x \geq 0\}$ , Range:  $\{y \geq 1\}$

☒ C. Domain:  $\{x \geq 1\}$ , Range:  $\{y \geq 0\}$

D. Domain:  $\{x \geq 1\}$ , Range:  $\{y \geq 1\}$

$$D_f: x \geq 0 \quad R_f: y \geq 1$$

$$D_{f^{-1}}: x \geq 1 \quad R_{f^{-1}}: y \geq 0$$

2. The acute angle (to the nearest degree) between the vectors  $\underline{u} = \begin{pmatrix} \sqrt{3} \\ 2 \end{pmatrix}$  and  $\underline{v} = \begin{pmatrix} 9 \\ \sqrt{17} \end{pmatrix}$  is:

☒ A.  $24^\circ$

B.  $51^\circ$

C.  $46^\circ$

D.  $48^\circ$

$$\begin{aligned} \cos \theta &= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \\ &= \frac{(9\sqrt{3} + 2\sqrt{17})}{\sqrt{11} \times \sqrt{98}} \\ \therefore \theta &\approx 24^\circ 29' 34'' \text{ closer to } 24^\circ. \end{aligned}$$

3. The solution to the inequality  $\frac{3}{x-2} \leq 4$  is given by:

A.  $x < -2$  and  $x \geq -\frac{11}{4}$

B.  $x > -2$  and  $x \leq -\frac{11}{4}$

☒ C.  $x < 2$  and  $x \geq \frac{11}{4}$

D.  $x > 2$  and  $x \leq \frac{11}{4}$

$$\begin{aligned} 3(x-2) &\leq 4(x-2)^2, \quad x \neq 2 \\ 4(x-2)^2 &\geq 3(x-2) \\ 4(x-2)^2 - 3(x-2) &\geq 0 \\ (x-2)(4x-8-3) &\geq 0 \\ (x-2)(4x-11) &\geq 0 \end{aligned}$$



4. A function is defined as  $f(x) = \tan^{-1}(\tan(x))$ .

The value of  $f\left(\frac{9\pi}{4}\right)$  is:

- A.  $\frac{\pi}{4}$
- B.  $\frac{5\pi}{4}$
- C.  $\frac{7\pi}{4}$
- D.  $\frac{9\pi}{4}$

$$\begin{aligned} f\left(\frac{9\pi}{4}\right) &= \tan^{-1}\left(\tan\left(\frac{9\pi}{4}\right)\right) \\ &= \tan^{-1}\left(\tan\left(2\pi + \frac{\pi}{4}\right)\right) \\ &= \tan^{-1}\left(\tan\frac{\pi}{4}\right) \\ &= \frac{\pi}{4} \end{aligned}$$

5. When the polynomial  $P(x)$  is divided by  $x+1$  it gives a remainder of  $n^2$ , where  $n$  is a positive integer. When  $P(x)$  is divided by  $(x+1)(x+3)$  it gives a remainder  $(nx+6)$ . The value of  $n$  is:

- A. 1
- B. 3
- C. 6
- D. 2

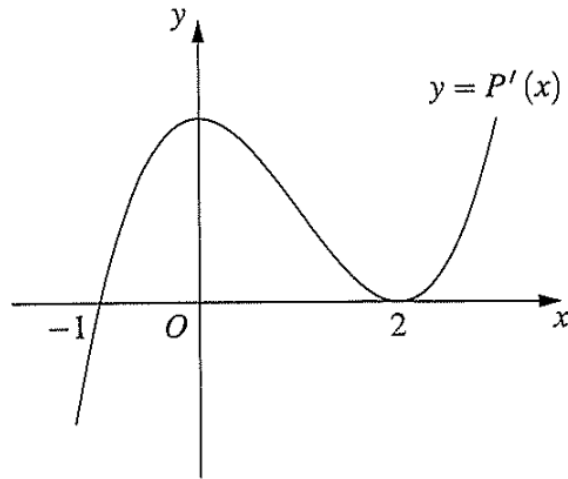
$$\begin{aligned} P(-1) &= n^2 \\ P(x) &= (x+1)(x+3) + (nx+6) \\ P(-1) &= -n+6 \\ n^2 &= -n+6 \\ n^2+n-6 &= 0 \\ (n+3)(n-2) &= 0 \therefore n=2, \text{ given } n = \text{positive integer} \end{aligned}$$

6. The  $\int \sin 3x \sin x \, dx$  is equal to:

- A.  $\frac{1}{4} \sin 2x - \frac{1}{8} \sin 4x + C$
- B.  $\frac{1}{8} \sin 4x - \frac{1}{4} \sin 2x + C$
- C.  $\frac{1}{4} \cos 2x - \frac{1}{8} \cos 4x + C$
- D.  $\frac{1}{8} \cos 4x - \frac{1}{4} \cos 2x + C$

$$\begin{aligned} \int \sin 3x \sin x \, dx &= \frac{1}{2} \int (\cos 2x - \cos 4x) \, dx \\ &= \frac{1}{2} \left[ \frac{1}{2} \sin 2x - \frac{1}{4} \sin 4x \right] + C \\ &= \frac{1}{4} \sin 2x - \frac{1}{8} \sin 4x + C \end{aligned}$$

7. In the graph below,  $y = P'(x)$  represents the first derivative of a polynomial  $P(x)$  of degree 4, where  $P(x)$  has a multiple root.



Which of the following could be true about the polynomial  $P(x)$  ?

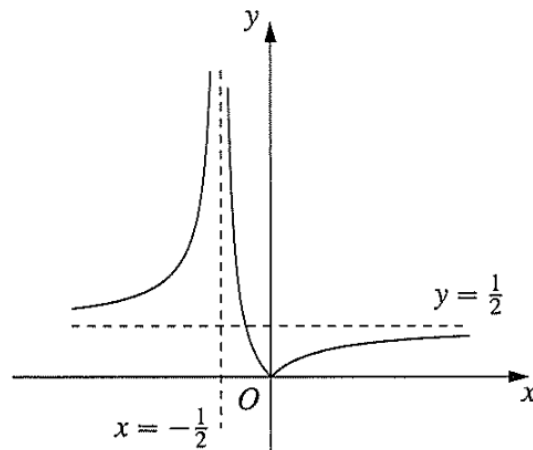
- A.  $x = -1$  is a root of multiplicity 3.
- B.  $x = 0$  is a root of multiplicity 2.
- C.  $x = 2$  is a root of multiplicity 2.
- ☒ D.  $x = 2$  is a root of multiplicity 3.

8. The derivative of  $f(x) = 2x^2 \cos^{-1}(2x)$  is:

- A.  $\frac{-8x}{\sqrt{1-2x^2}}$
- B.  $\frac{-8x}{\sqrt{1-4x^2}}$
- C.  $\frac{-4x^2}{\sqrt{1-2x^2}} + 4x \cos^{-1} 2x$
- ☒ D.  $\frac{-4x^2}{\sqrt{1-4x^2}} + 4x \cos^{-1} 2x$

$$\begin{aligned} f'(x) &= [\cos^{-1}(2x)] 4x - 2x^2 \cdot \frac{2}{\sqrt{1-4x^2}} \\ &= -\frac{4x^2}{\sqrt{1-4x^2}} + 4x \cos^{-1}(2x) \end{aligned}$$

9. The diagram below shows the graph of the function  $y = g(x)$ , which is the result of a set of transformations on the graph of  $y = f(x)$ .



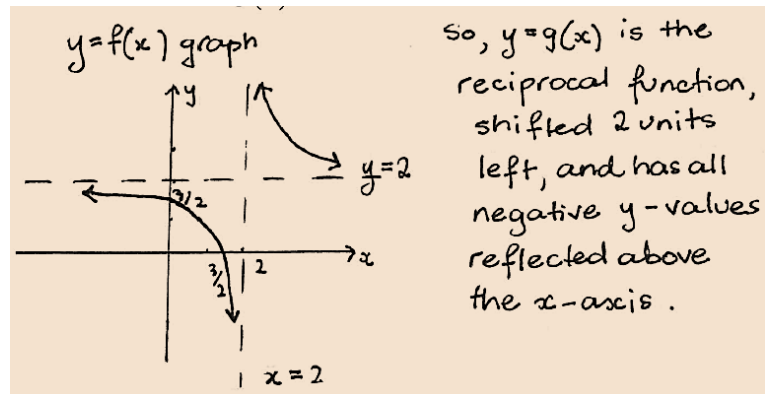
If  $f(x) = \frac{1}{x-2} + 2$ , which equation best represents  $g(x)$ ?

A.  $g(x) = \frac{1}{|f(x-2)|}$

B.  $g(x) = |f(x-2)|$

☒ C.  $g(x) = \frac{1}{|f(x+2)|}$

D.  $g(x) = |f(x+2)|$



10. Vector  $\underline{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$  is projected on to vector  $\underline{b} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$ . Given that  $\text{Proj}_{\underline{b}} \underline{a} = \frac{1}{2} \underline{b}$ , which of the following could possibly be vector  $\underline{a}$ ?

A.  $\underline{a} = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$

B.  $\underline{a} = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$

☒ C.  $\underline{a} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$

D.  $\underline{a} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

Handwritten solution for Question 10:

$$\frac{1}{2} \underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \times \underline{b}$$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \frac{4a_1 + 6a_2}{52} \times \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \frac{2a_1 + 3a_2}{26} \times \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \frac{10 + 3}{13} \times \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad (\text{substitute answers})$$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \text{'C' works}$$

## Section II

60 marks

Attempt Questions 11 – 15

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 11 – 15, your responses should include relevant mathematical reasoning and/or calculations.

**Question 11** (13 marks) Use the Question 11 Writing Booklet

- (a) Let  $P$  be the point  $(12, 5)$ . Find the position vector of point  $P$ , and hence find a unit vector in the direction of the position vector.

2

(a)  $P(12, 5)$ , position vector  $12\mathbf{i} + 5\mathbf{j}$  ✓ (1 for vector)

Unit vector  $= \frac{\vec{OP}}{|\vec{OP}|}$

$= \frac{12\mathbf{i} + 5\mathbf{j}}{\sqrt{144 + 25}}$

$= \frac{1}{13}(12\mathbf{i} + 5\mathbf{j})$

1 for either answer

- (b) Find the exact value of  $\int \sin^2 \frac{2}{3}x \, dx$ .

1

$\int \sin^2\left(\frac{2}{3}x\right) dx$

$= \frac{1}{2} \int (1 - \cos \frac{4}{3}x) dx$

$= \frac{1}{2} \left[ x - \frac{3}{4} \sin \frac{4}{3}x \right] + C$  ✓ (1 for correct primitive function)

Do not deduct mark if C is missing

(c) Use the  $t$ -identities to solve the equation  $2\sin x + \cos x + 1 = 0$  for  $0 \leq x \leq 2\pi$ .

4

(c)  $2\sin x + \cos x + 1 = 0$

$$\frac{4t}{1+t^2} + \frac{1-t^2}{1+t^2} + 1 = 0$$

$$\frac{4t + 1 - t^2 + 1 + t^2}{1+t^2} = 0$$

$$\frac{4t + 2}{1+t^2} = 0$$

① for some progress made after substituting  $t$  results

$$4t + 2 = 0$$

$$\therefore t = -\frac{1}{2}$$

① for  $t$  value

$$\tan \frac{x}{2} = -\frac{1}{2}, \quad 0 \leq \frac{x}{2} \leq \pi$$

$$\text{related } \angle \doteq 0.4636^\circ$$

$$(\text{accept related } \angle = \tan^{-1}(\frac{1}{2}))$$

$$\therefore \frac{x}{2} = \pi - 0.4636^\circ \text{ or } \frac{x}{2} = \pi - \tan^{-1}(\frac{1}{2})$$

$$x = 2(\pi - 0.4636^\circ)$$

$$x \doteq 5.3559^\circ$$

or accept

$$x = 2(\pi - \tan^{-1}(\frac{1}{2}))$$

Test  $x = \pi$

$$2\sin\pi + \cos\pi + 1 = 0$$

$$0 - 1 + 1 = 0$$

$$0 = 0$$

$$\therefore x = \pi, 5.3559^\circ (4 \text{ d.p.})$$

$$2(\pi - \tan^{-1} \frac{1}{2})$$

② marks

One for each answer.

(d) Find  $\int \frac{1}{\sqrt{64-49x^2}} dx$ .

2

(d)  $\int \frac{1}{\sqrt{64-49x^2}} dx$   
 $= \int \frac{1}{\sqrt{8^2-(7x)^2}}$   
 $= \frac{1}{7} \int \frac{7}{\sqrt{8^2-(7x)^2}} \quad \checkmark \quad \textcircled{1} \text{ for showing working}$   
 $= \frac{1}{7} \sin^{-1}\left(\frac{7x}{8}\right) + C \quad \checkmark \quad \textcircled{1} \text{ correct primitive function}$

(e) (i) Express  $\cos x - \sqrt{3} \sin x$  in the form  $R \cos(x+\alpha)$  for  $R > 0$  and  $\alpha$  acute.

2

$\cos x - \sqrt{3} \sin x = R \cos(x+\alpha)$   
 $= R [\cos x \cos \alpha - \sin x \sin \alpha]$   
 $\cos x - \sqrt{3} \sin x = R \cos x \cos \alpha - R \sin x \sin \alpha$   
 $R \cos \alpha = 1, \quad R \sin \alpha = \sqrt{3}$   
 $R^2 (\cos^2 \alpha + \sin^2 \alpha) = 1 + 3$   
 $R^2 (1) = 4$   
 $\therefore R = 2 \quad (R > 0)$  }  $\checkmark \quad \textcircled{1} \text{ for } R \text{ with working}$   
 $\therefore \frac{R \sin \alpha}{R \cos \alpha} = \frac{\sqrt{3}}{1}$   
 $\tan \alpha = \sqrt{3}$   
 $\therefore \alpha = \frac{\pi}{3}, \quad \alpha = \text{acute } \angle \quad \checkmark \quad \textcircled{1} \text{ for } \alpha \text{ with working}$   
 $\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right)$

(ii) Hence, solve  $\cos x - \sqrt{3} \sin x = 2$  for  $0 \leq x \leq 2\pi$ .

2

f) ii)  $\cos x - \sqrt{3} \sin x = 2$ ,  $0 \leq x \leq 2\pi$

① for  $\checkmark$  trig equation rewritten in alternate form

$$2 \cos\left(x + \frac{\pi}{3}\right) = 2$$
$$\cos\left(x + \frac{\pi}{3}\right) = 1, \quad \frac{\pi}{3} \leq x + \frac{\pi}{3} \leq \frac{7\pi}{3}$$
$$x + \frac{\pi}{3} = 2\pi$$
$$\therefore x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3} \quad \checkmark$$

① for solution

End of Question 11

**Question 12** (12 marks) Use the Question 12 Writing Booklet

- (a) There are four teams consisting of two tennis players in each team.

2

They are seated around a circular table.

In how many ways can the teams be arranged around the table if the players in each team are to stay together.

Number of ways

$$= (4-1)! \times (2!)^4$$
$$= 3! \cdot (2!)^4 \text{ or } 96$$

① for number of ways around a circle

① for arranging pairs  $(2!)^4$

Accept any version for answer

- (b) By using the substitution  $x = \sin \theta$ , find  $\int_0^{\frac{1}{2}} (1-x^2)^{-\frac{3}{2}} dx$

3

$x = \sin \theta$ ,  $dx = \cos \theta d\theta$

$x = 0 \rightarrow \theta = 0$

$x = \frac{1}{2} \rightarrow \theta = \frac{\pi}{6}$

$$\int_0^{\frac{1}{2}} (1-x^2)^{-\frac{3}{2}} dx$$
$$= \int_0^{\frac{\pi}{6}} (1-\sin^2 \theta)^{-\frac{3}{2}} \cos \theta d\theta$$

① rewriting integral correctly in terms of  $\theta$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{(\sqrt{\cos^2 \theta})^3} \cos \theta d\theta$$
$$= \int_0^{\frac{\pi}{6}} \frac{1}{\cos^3 \theta} \cdot \cos \theta d\theta$$
$$= \int_0^{\frac{\pi}{6}} \sec^2 \theta d\theta$$

① for progress

$$= [\tan \theta]_0^{\frac{\pi}{6}}$$
$$= \tan \frac{\pi}{6} - \tan 0$$
$$= \frac{1}{\sqrt{3}}$$

① for answer

(c) Prove by mathematical induction that  $3^{3n-1} + 5^{3n-2}$  is divisible by 7,

4

for any integer  $n \geq 1$ .

(G) Test  $n=1$

$$3^{3n-1} + 5^{3n-2}$$

$$= 3^2 + 5^1$$

$= 14$  which is divisible by 7 ✓

∴ statement is true for  $n=1$

① for testing  
 $n=1$

Assume true for  $n=k$ ,  $k = +ve$  integer

$$\text{i.e. } 3^{3k-1} + 5^{3k-2} = 7P, P = \text{some integer}$$

And prove true for  $n=k+1$

$$3^{3(k+1)-1} + 5^{3(k+1)-2}$$

$$= 3^{3k+2} + 5^{3k+1}$$

$$= 3^{3k-1+3} + 5^{3k+1}$$

$$= (7P - 5^{3k-2}) 3^3 + 5^{3k+1} \text{ using assumption } \checkmark$$

① for using assumption

$$= 27(7P - 5^{3k-2}) + 5^{3k-2+3}$$

$$= 189P - 27(5^{3k-2}) + 5^{3k-2}(5^3) \checkmark$$

① For correct progress

$$= 189P + 98(5^{3k-2})$$

$$= 7[27P + 14(5^{3k-2})]$$

which is divisible by 7

∴ statement is true for  $n=k+1$ ,

assuming true for  $n=k$

① for showing divisible by 7 when  $n=k+1$

So by mathematical induction,  
statement is true for all integers  
 $n \geq 1$ . ✓

(d) A cake tin cools such that its temperature  $T$  degrees,  $t$  minutes after it is removed

3

from the oven is given by:  $T = R + Ae^{-kt}$  Where  $k$ ,  $R$  and  $A$  are constants

It is known that:  $\frac{dT}{dt} = -k(T - R)$  Do NOT prove this.

When removed from an oven the cake tin has a temperature of  $180^\circ\text{C}$ . If the cake tin takes one minute to cool to  $150^\circ\text{C}$  and the room temperature is  $20^\circ\text{C}$ , find the time to the nearest minute it takes for the cake tin to cool to  $80^\circ\text{C}$ . (Assume that the cake tin cools at a rate proportional to the difference between the temperature of the cake tin and the temperature of the surrounding air.)

(d)  $T = R + Ae^{-kt}$ ,  $\frac{dT}{dt} = -k(T - R)$

at  $t = 0$ ,  $T = 180^\circ\text{C}$  ①  
 $t = 1$ ,  $T = 150^\circ\text{C}$  ②  
 $t = ?$ ,  $T = 80^\circ\text{C}$  ③

$T = R + Ae^{-kt}$ ,  $R = 20$   
 $T = 20 + Ae^{-kt}$  (sub R)  
 $180 = 20 + Ae^0$  (sub ①)  
 $160 = A$  ✓ ① for A  
 $\therefore T = 20 + 160e^{-kt}$   
 $150 = 20 + 160e^{-k}$  (sub ②)  
 $130 = 160e^{-k}$

$e^{-k} = \frac{13}{16}$

$\log_e\left(\frac{13}{16}\right) = -k$   
 $\therefore k = -\ln\left(\frac{13}{16}\right) = \ln\left(\frac{16}{13}\right)$  ✓ ① for k

$\therefore T = 20 + 160e^{\left[\ln\left(\frac{16}{13}\right)t\right]}$

$80 = 20 + 160e^{\left[\ln\left(\frac{16}{13}\right)t\right]}$  sub ③  
 $60 = 160e^{\left[\ln\left(\frac{16}{13}\right)t\right]}$   
 $\frac{3}{8} = e^{\left[\ln\left(\frac{16}{13}\right)t\right]}$

$\therefore \ln\left(\frac{3}{8}\right) = \ln\left(\frac{16}{13}\right) \cdot t$

$\therefore t = \frac{\ln\left(\frac{3}{8}\right)}{\ln\left(\frac{16}{13}\right)}$

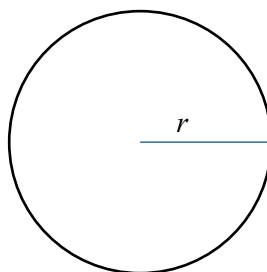
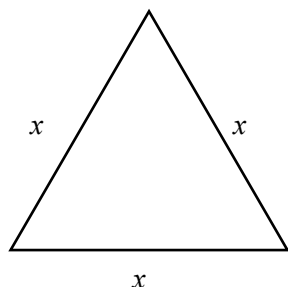
$\approx 4.7237$  ✓ ① for correct solution. Ignore rounding errors but make a comment

$\therefore$  It takes approx. 5 minutes for cake tin to cool to  $80^\circ\text{C}$ .

End of Question 12

**Question 13** (13 marks) Use the Question 13 Writing Booklet

- (a) The side lengths of an equilateral triangle are shrinking at a rate of  $\sqrt{x} \text{ m/s}$ , where  $x$  metres is the length of each side of the triangle. The triangle and the circle shown below always have the same area  $A$ .



- (i) Show that the rate of change of the area of the triangle is:

2

$$\frac{-\sqrt{3}x\sqrt{x}}{2} \text{ m}^2/\text{s}$$

(a) i)  $A = \frac{1}{2} \times x \times x \times \sin 60^\circ$

$$= \frac{1}{2} \times x^2 \times \frac{\sqrt{3}}{2}$$
$$A = \frac{\sqrt{3}}{4} x^2 \quad \checkmark \quad \textcircled{1} \text{ for area of triangle}$$
$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$
$$= \frac{\frac{1}{2} \sqrt{3} x}{\cancel{2}} \times -\sqrt{x} \quad \checkmark \quad \textcircled{1} \text{ for working}$$
$$= \frac{-\sqrt{3} x \sqrt{x}}{2}$$

(ii) Find the rate of change of the radius of the circle when  $x = 5$  metres.

3

Give the answer correct to 2 decimal places.

ii) Circle:  $\frac{dr}{dt} = ?$  when  $x = 5$

$$A = \pi r^2$$

$$\frac{\sqrt{3}}{4} x^2 = \pi r^2$$

$$r = \sqrt{\frac{\sqrt{3} x^2}{4\pi}} = \left( \frac{\sqrt{3} x^2}{4\pi} \right)^{1/2}$$

$$r = \frac{3^{1/4}}{2\sqrt{\pi}} x$$

① for attempting to find  $r$

$$\frac{dr}{dt} = \frac{dr}{dx} \times \frac{dx}{dt}$$

$$= \frac{3^{1/4}}{2\sqrt{\pi}} \times -\sqrt{x}$$

① for working to find  $\frac{dr}{dt}$

$$\text{at } x=5 \quad \frac{dr}{dt} = \frac{3^{1/4}}{2\sqrt{\pi}} \times -\sqrt{5}$$

$$= -0.83 \text{ m/s (2 d.p.)}$$

✓ ① for answer. If left exact, ignore & make comment

$$\frac{dr}{dt} = ? \text{ when } x=5$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$\frac{-\sqrt{3} x \sqrt{x}}{2} = 2\pi r \times \frac{dr}{dt}$$

$$\frac{-\sqrt{3} x^{1.5}}{2} = \frac{2 \times \pi \times 3^{1/4} x}{2\sqrt{\pi}} \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{-\sqrt{3} x^{1.5} \times 2 \times \sqrt{\pi}}{2 \times 2 \times \pi \times 3^{1/4} \times 5} \quad x=5$$

$$= -0.83 \text{ m/s (2 d.p.)}$$

(b) By considering  $(1+x)^{2n} + (1-x)^{2n}$ , where  $n$  is a positive integer, show that:

3

$$1 + \binom{2n}{2} + \binom{2n}{4} + \dots + \binom{2n}{2n} = \frac{4^n}{2}.$$

(b)  $(1+x)^{2n} + (1-x)^{2n}$

$$= \binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \binom{2n}{3}x^3 + \dots + \binom{2n}{2n}x^{2n}$$

$$+ \left[ \binom{2n}{0} - \binom{2n}{1}x + \binom{2n}{2}x^2 - \binom{2n}{3}x^3 + \dots + \binom{2n}{2n}x^{2n} \right]$$

① for expanding correctly

$$= 2 \left[ \binom{2n}{0} + \binom{2n}{2}x^2 + \binom{2n}{4}x^4 + \dots + \binom{2n}{2n}x^{2n} \right]$$

Let  $x=1$ ;

① for substituting  $x=1$

$$(1+1)^{2n} + (1-1)^{2n} = 2 \left[ \binom{2n}{0} + \binom{2n}{2}1^2 + \binom{2n}{4}1^4 + \dots + \binom{2n}{2n}1^{2n} \right]$$

$$2^{2n} = 2 \left[ 1 + \binom{2n}{2} + \binom{2n}{4} + \binom{2n}{6} + \dots + \binom{2n}{2n} \right]$$

$$\therefore 1 + \binom{2n}{2} + \binom{2n}{4} + \dots + \binom{2n}{2n} = \frac{2^{2n}}{2} = \frac{4^n}{2}$$

① for working giving required answer

(c)  $OABC$  is trapezium with  $\overrightarrow{OA} = 3\overrightarrow{CB}$ .  $\overrightarrow{OA} = \underline{a}$  and  $\overrightarrow{OC} = \underline{c}$ .

$E$  and  $F$  are midpoints of  $OB$  and  $CA$  respectively.

(i) Show that  $\overrightarrow{OE} = \frac{1}{2}\left(\underline{c} + \frac{1}{3}\underline{a}\right)$ .

1

(i) Given:  $\overrightarrow{OA} = 3\overrightarrow{CB}$ ,  $\overrightarrow{OA} = \underline{a}$ ,  $\overrightarrow{OC} = \underline{c}$   
 $E, F$  are midpoints of  $OB, CA$

Show  $\overrightarrow{OE} = \frac{1}{2}\left(\underline{c} + \frac{1}{3}\underline{a}\right)$

$\overrightarrow{OB} = \overrightarrow{OC} + \overrightarrow{CB}$   
 $= \underline{c} + \frac{1}{3}\underline{a}$  ✓

$\therefore \overrightarrow{OE} = \frac{1}{2}\left(\underline{c} + \frac{1}{3}\underline{a}\right)$  given  $OE = \frac{1}{2}OB$

① mark for correct working giving required answer

(ii) Find  $\overrightarrow{EF}$ .

3

ii)  $\overrightarrow{EF} = \overrightarrow{CF} - \overrightarrow{CE}$  where  $\overrightarrow{CF} = \frac{1}{2}(\overrightarrow{OA} - \overrightarrow{OC})$

$\overrightarrow{EF} = \frac{1}{2}(\underline{a} - \underline{c}) - \left(\frac{1}{6}\underline{a} - \frac{1}{2}\underline{c}\right) = \frac{1}{2}(\underline{a} - \underline{c})$  ✓ ① mark for  $\overrightarrow{CF}$

$= \frac{1}{2}\underline{a} - \frac{1}{2}\underline{c} - \frac{1}{6}\underline{a} + \frac{1}{2}\underline{c}$   $\overrightarrow{CE} = \overrightarrow{OE} - \overrightarrow{OC}$

$= \frac{2}{6}\underline{a}$   $= \frac{1}{2}\left(\underline{c} + \frac{1}{3}\underline{a}\right) - \underline{c}$

$= \frac{1}{3}\underline{a}$  ✓ ① mark for  $\overrightarrow{CE}$

① mark for  $\overrightarrow{EF} = \frac{1}{3}\underline{a}$

(iii) Hence, prove that  $CEFB$  is a parallelogram.

1

iii)  $\overrightarrow{CB} = \frac{1}{3}\overrightarrow{OA}$  and  $\overrightarrow{EF} = \frac{1}{3}\underline{a}$  (from ii)

$= \frac{1}{3}\underline{a}$  (given)

$\therefore \overrightarrow{CB}$  is parallel to  $\overrightarrow{EF}$

and  $|\overrightarrow{EF}| = |\overrightarrow{CB}|$

$\therefore CEFB$  is a parallelogram. ✓ ① mark for explanation

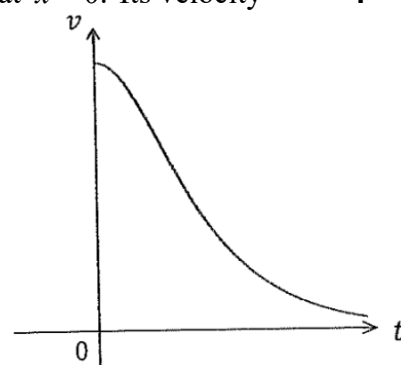
**Question 14** (11 marks) Use the Question 14 Writing Booklet

- (a) A particle  $P$  is moving along the  $x$  axis. At time  $t = 0$ , it was at  $x = 0$ . Its velocity

4

$$v \text{ ms}^{-1} \text{ at a time } t \text{ is given by } v = \frac{e^t}{1 + (e^t)^2}.$$

The graph below shows its velocity as a function of time.



Given that the displacement of the particle when  $t = k$  seconds is  $\tan^{-1} \frac{1}{2}$  metres, show that  $k = \ln 3$ .

(a) At  $t=0$ ,  $x=0$ . Also,  $x = \tan^{-1} \frac{1}{2}$  when  $t=k$

(4 marks)

$$v = \frac{e^t}{1 + (e^t)^2}$$

$$x = \int \frac{e^t}{1 + (e^t)^2} dt$$

$$x = \tan^{-1}(e^t) + C$$

① for correct integration

at  $t=0$ ,  $x=0 \therefore$

$$0 = \tan^{-1}(e^0) + C$$

$$0 = \tan^{-1}(1) + C$$

$$0 = \frac{\pi}{4} + C$$

$$\therefore C = -\frac{\pi}{4}$$

① for correct 'C'

$$\therefore x = \tan^{-1}(e^t) - \frac{\pi}{4}$$

Now:  $\tan^{-1}(\frac{1}{2}) = \tan^{-1}(e^k) - \frac{\pi}{4}$

$$\tan^{-1}(e^k) - \tan^{-1}(\frac{1}{2}) = \frac{\pi}{4}$$

$$\tan[\tan^{-1}(e^k) - \tan^{-1}(\frac{1}{2})] = \tan \frac{\pi}{4}$$

① for progress

using  $\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \tan \frac{\pi}{4}$

$$\frac{e^k - \frac{1}{2}}{1 + \frac{e^k}{2}} = 1$$

$$\frac{2e^k - 1}{2 + e^k} = 1$$

$$2e^k - 1 = 2 + e^k$$

$$e^k = 3$$

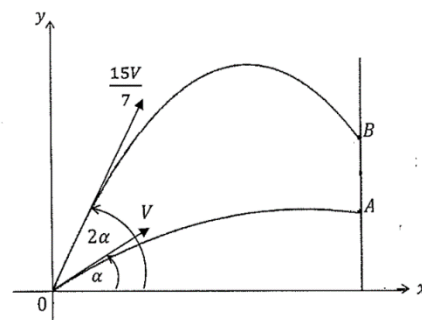
① for correct working giving  $e^k = 3$

i.e.  $\log_e 3 = k$

i.e.  $k = \ln 3$ .

(b) Jacob hits two balls at the same time from the origin  $O$  towards a target on a vertical wall.

The velocity of the first ball is  $V \text{ ms}^{-1}$  and its angle of projection is  $\alpha$  to the horizontal while the velocity of the second ball is  $\frac{15V}{7} \text{ ms}^{-1}$  and its angle of projection to the horizontal is  $2\alpha$ .



- (i) Show that the position vector of the second ball  $t$  seconds after be projected is given by: 3

$$\underline{r}_2(t) = \begin{pmatrix} \frac{15V}{7}t \cos 2\alpha \\ -\frac{1}{2}gt^2 + \frac{15V}{7}t \sin 2\alpha \end{pmatrix}$$

i) 2nd ball: (3 marks)

|                                                                                                                                        |                                                                            |                                 |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------|
| Horizontally<br>$\ddot{x} = 0 \hat{i}$                                                                                                 | Vertically<br>$\ddot{y} = -g \hat{j}$ ✓                                    | ① for acceleration vectors      |
| $\dot{x} = c_1 \hat{i}$                                                                                                                | $\dot{y} = -gt \hat{j} + c_2 \hat{j}$                                      |                                 |
| at $t=0$ , $\dot{x} = \frac{15V}{7} \cos 2\alpha \hat{i}$                                                                              | at $t=0$ , $\dot{y} = \frac{15V}{7} \sin 2\alpha \hat{j}$                  |                                 |
| $\therefore c_1 = \frac{15V}{7} \cos 2\alpha \hat{i}$                                                                                  | $c_2 = \frac{15V}{7} \sin 2\alpha \hat{j}$                                 | ① for deriving velocity vectors |
| $\therefore \dot{x} = \frac{15V}{7} \cos 2\alpha \hat{i}$                                                                              | $\therefore \dot{y} = (-gt + \frac{15V}{7} \sin 2\alpha) \hat{j}$ ✓        |                                 |
| $x = \frac{15Vt}{7} \cos 2\alpha \hat{i} + c_3$                                                                                        | $y = (-\frac{gt^2}{2} + \frac{15Vt}{7} \sin 2\alpha) \hat{j} + c_4$        |                                 |
| at $t=0$ , $x=0$ , $\therefore c_3=0$                                                                                                  | at $t=0$ , $y=0$ $\therefore c_4=0$                                        |                                 |
| $\therefore x = \frac{15Vt}{7} \cos 2\alpha \hat{i}$                                                                                   | $\therefore y = (-\frac{gt^2}{2} + \frac{15Vt}{7} \sin 2\alpha) \hat{j}$ ✓ |                                 |
| i.e. $\underline{r}_2(t) = \begin{bmatrix} \frac{15Vt}{7} \cos 2\alpha \\ -\frac{gt^2}{2} + \frac{15Vt}{7} \sin 2\alpha \end{bmatrix}$ |                                                                            | ① for deriving position vectors |

(ii) It is known that both balls hit the wall after  $T$  seconds. Show that  $\cos \alpha = \frac{5}{6}$ .

4

Ball 1:

Considering only horizontal motion:

$$\ddot{x} = 0$$

$$\dot{x} = c_5$$

at  $t=0$ ,  $\dot{x} = V \cos \alpha$

$\therefore c_5 = V \cos \alpha$

$\therefore \dot{x} = V \cos \alpha$

$x = Vt \cos \alpha + c_6$

at  $t=0$ ,  $x=0$   $\therefore c_6 = 0$

$\therefore x = Vt \cos \alpha$  ✓

① for correct horizontal displacement vector for ball. Student does not need to derive this. Can just say: 'similarly  $x = vt \cos \alpha$  for ball.'

After  $T$  seconds:

$$VT \cos \alpha = \frac{15V}{7} T \cos 2\alpha$$

① for equating horizontal displacements

$$7VT \cos \alpha = 15VT (2 \cos^2 \alpha - 1)$$

$$7 \cos \alpha = 30 \cos^2 \alpha - 15$$

$$30 \cos^2 \alpha - 7 \cos \alpha - 15 = 0$$

① for equation in terms of  $\cos \alpha$

$$\cos \alpha = \frac{7 \pm \sqrt{49 - 4(30)(-15)}}{60}$$

$$\cos \alpha = \frac{7 \pm 43}{60}$$

$$\cos \alpha = \frac{50}{60}, \cos \alpha > 0$$

$$\therefore \cos \alpha = \frac{5}{6}$$

✓ ① for solving for  $\cos \alpha = \frac{5}{6}$  must show solving steps

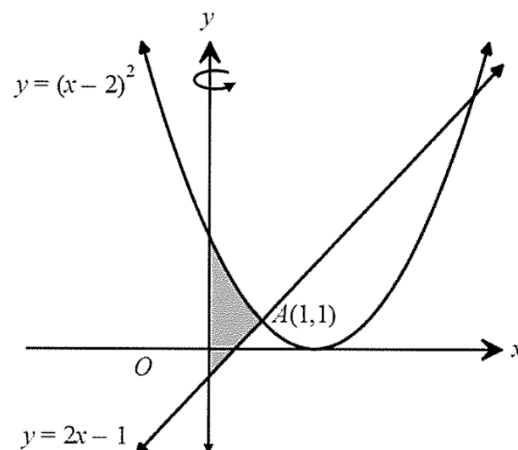
End of Question 14

**Question 15** (11 marks) Use the Question 15 Writing Booklet

- (a) Consider the shaded region bounded by the line  $y = 2x - 1$ , the curve  $y = (x - 2)^2$  and the  $y$ -axis.

4

Find the volume of the solid of revolution formed by rotating this region about the  $y$ -axis.



(a) Divide the shaded region into two parts through the line  $y = 1$ . When the region below this line is rotated about the  $y$ -axis we obtain a cone of radius 1 and height 2.

$$\begin{aligned} \therefore V &= \frac{1}{3} \pi (1)^2 (2) + \pi \int_1^4 x^2 dy \\ &= \frac{2\pi}{3} + \pi \int_1^4 (4 + y - 4\sqrt{y}) dy \quad \text{① for integral that gives volume} \\ &= \frac{2\pi}{3} + \pi \left[ 4y + \frac{y^2}{2} - 4 \cdot \frac{2}{3} y^{3/2} \right]_1^4 \quad \text{For the shaded area,} \\ &= \frac{2\pi}{3} + \pi \left[ \left( 16 + 8 - \frac{64}{3} \right) - \left( 4 + \frac{1}{2} - \frac{8}{3} \right) \right] \quad \text{① for integrating correctly} \\ &= \frac{2\pi}{3} + \pi \left[ \frac{8\pi}{3} - \frac{11\pi}{6} \right] \quad \text{① for substitution} \\ &= \frac{2\pi}{3} + \frac{5\pi}{6} \\ &= \frac{3\pi}{2} \text{ units}^3 \quad \text{① for answer} \end{aligned}$$

(or)

$$\begin{aligned} V &= \pi \int_{-1}^1 (y+1)^2 dy + \pi \int_1^4 (4+y-4\sqrt{y}) dy \quad y=2x-1 \rightarrow x=\frac{y+1}{2} \\ &= \pi \left[ \frac{(y+1)^3}{3} \right]_{-1}^1 + \pi \left[ 4y + \frac{y^2}{2} - 4 \cdot \frac{2}{3} y^{3/2} \right]_1^4 \\ &= \pi \left[ \frac{8}{3} - 0 \right] + \pi \left[ \left( 16 + 8 - \frac{64}{3} \right) - \left( 4 + \frac{1}{2} - \frac{8}{3} \right) \right] \\ &= \frac{2\pi}{3} + \frac{5\pi}{6} \\ &= \frac{3\pi}{2} \text{ units}^3 \end{aligned}$$

(b) (i) Show that the solution of  $\cos 3x - \sin 2x = 0$ , for  $0 < x < \frac{\pi}{2}$  is given by

3

$$\sin x = \frac{\sqrt{5}-1}{4}.$$

You may use the identity  $\cos 3x = 4\cos^3 x - 3\cos x$ . DO NOT PROVE THIS.

b) i)  $\cos 3x - \sin 2x = 0, 0 < x < \frac{\pi}{2}$

$$4\cos^3 x - 3\cos x - \sin 2x = 0$$

$$4\cos^3 x - 3\cos x - 2\sin x \cos x = 0$$

$$\cos x (4\cos^2 x - 2\sin x - 3) = 0$$

$$\cos x (4(1 - \sin^2 x) - 2\sin x - 3) = 0$$

$$-\cos x (4\sin^2 x + 2\sin x - 1) = 0$$

✓ ① for progress

Now  $\cos x = 0$  when  $x = \frac{\pi}{2}$ , which

is not in domain given,  $0 < x < \frac{\pi}{2}$

✓ ① for why  $x \neq \frac{\pi}{2}$

$$4\sin^2 x + 2\sin x - 1 = 0$$

$$\sin x = \frac{-2 \pm \sqrt{4 - 4(4)(-1)}}{8}$$

$$\sin x = \frac{-2 \pm \sqrt{20}}{8} \quad (\sqrt{20} = 2\sqrt{5})$$

$$\sin x = \frac{-1 \pm \sqrt{5}}{4}$$

$$\sin x = \frac{\sqrt{5}-1}{4} \quad \text{as } \sin x > 0$$

in domain  
 $0 < x < \frac{\pi}{2}$

① for working giving required value for  $\sin x$ . Do not deduct marks twice for not stating restrictions

(ii) Without a calculator, verify that  $x = \frac{\pi}{10}$  is a solution to  $\cos 3x = \sin 2x$ .

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b) i)  $\cos 3x = \sin 2x$

If  $x = \frac{\pi}{10}$ ,  $\cos 3x$

$= \cos \frac{3\pi}{10}$

$= \sin\left(\frac{\pi}{2} - \frac{3\pi}{10}\right)$  ✓ ① mark  
'show' question

$= \sin \frac{2\pi}{10}$

∴  $x = \frac{\pi}{10}$  is a solution

to  $\cos 3x = \sin 2x$

(iii) Using the results obtained in parts (i) and (ii) prove that  $\sin \frac{\pi}{5} \cos \frac{\pi}{10} = \frac{\sqrt{5}}{4}$ .

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### END OF EXAMINATION

(iii)  $\sin \frac{\pi}{5} \cos \frac{\pi}{10}$

$= \sin \frac{2\pi}{10} \cos \frac{\pi}{10}$

$= 2 \sin \frac{\pi}{10} \cos \frac{\pi}{10} \cos \frac{\pi}{10}$

$= 2 \sin \frac{\pi}{10} \cos^2 \frac{\pi}{10}$

either ✓ ① mark for recognising  
use of double angle formula

$= 2 \sin \frac{\pi}{10} (1 - \sin^2 \frac{\pi}{10})$

$= 2 \left( \frac{\sqrt{5}-1}{4} \right) \left( 1 - \left( \frac{\sqrt{5}-1}{4} \right)^2 \right)$  ✓ ①  
using (i) & (ii)

$= \left( \frac{\sqrt{5}-1}{2} \right) \left( \frac{16 - (5 - 2\sqrt{5} + 1)}{16} \right)$

$= \left( \frac{\sqrt{5}-1}{2} \right) \left( \frac{10 + 2\sqrt{5}}{16} \right)$

$= \left( \frac{\sqrt{5}-1}{2} \right) \left( \frac{5 + \sqrt{5}}{8} \right)$

$= \frac{5\sqrt{5} + 5 - 5 - \sqrt{5}}{16}$

$= \frac{4\sqrt{5}}{16}$

$= \frac{\sqrt{5}}{4}$

✓ ① mark for  
working leading  
to required answer.

i.e.  $\sin \frac{\pi}{5} \cos \frac{\pi}{10} = \frac{\sqrt{5}}{4}$